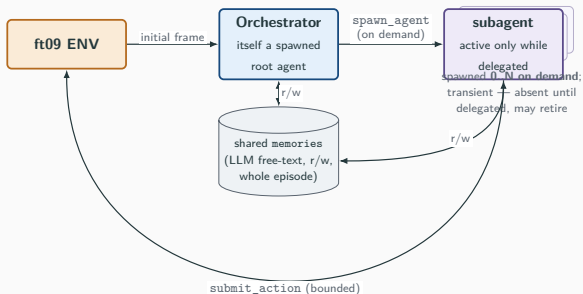


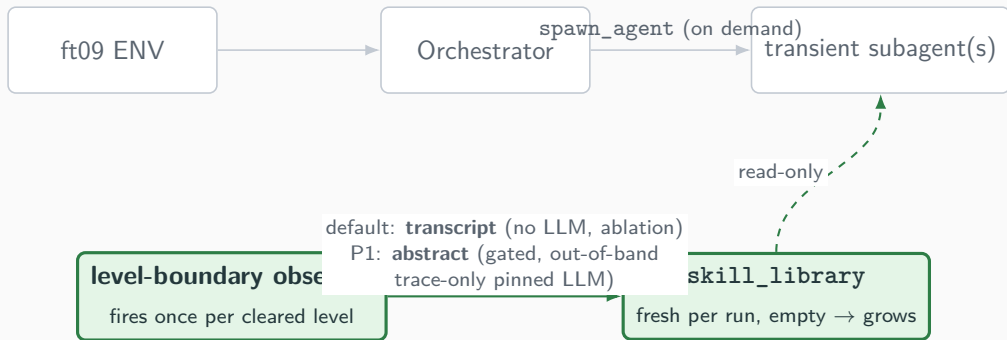
Original Symbolica: a spawned-root orchestrator that delegates on demand

infra (unchanged): LLM calls → agentic-server :2345 → TRAPI proxy :9091 (Bearer→AAD) →
agl-dev / TRAPI gpt-5.5 · no M1–M4 modules · per-turn CoT provider-encrypted



Byte-identical original Symbolica: one orchestrator (itself a spawned agent) dynamically spawns *transient* subagents *only when it delegates* — N varies, none always-running, *no fixed pipeline* — all

Trace2Skill: same loop + one additive read-only skill directory



Sole change: a fresh per-run, **read-only** `skill_library`. A level-boundary observer writes one entry per cleared level — **default** = verbatim *transcript* (deterministic, no LLM: the codex-required *ablation* arm); **P1** = gated *abstract* mode (out-of-band, trace-only, pinned LLM → mechanism/meta-reasoning skill). `none` runs zero of this; the agent only reads the directory.

Status: transcript signal is directional only; abstract P1 under test

Transcript mode = the ABLATION arm (not the thesis)

same-seed trace2skill $\geq$ none (ΔL)	5/5 directional
evidence quality	8/9 M3-censored
causal reliance on skill	not tested
no-skill is <i>not</i> generally stuck	none s43 $\rightarrow$ L6

Abstract mode = the actual thesis (P1)

estimand	trace-derived abstract skill induction + reuse
P1 code	9/9 offline tests
P1 paired experiment	running (early, M3-slow)
clean-rev4 win	no (not claimed)

Two byte-faithful L6 wins exist:

s47 trace2skill — L6 (101 turns)

s43 none (no skill) — L6 (118 turns)

So reaching L6 is not skill-exclusive; the transcript-mode ΔL is a directional, M3-censored signal — the abstract-mode causal test is what is now running.

Honest boundary: the only verified mechanistic fact is that the stored skill text re-enters the LLM body.input after consolidation (87/145, transcript mode); *causal* reliance and the abstract-mode effect are **not yet measured**.